PayrollDB

Database Schema Reference

7 tables **67** columns **21** indexes

Engine SQLite 3.39+ **License** MIT **Version** 1.0.0

Contents

Overview

PayrollDB is a relational database designed for managing employee compensation, payroll processing, performance evaluations, and bonus tracking. The schema is organized around a central employees table with six supporting tables covering departments, salary history, payroll records, deductions, performance reviews, and bonuses.

Key design features include generated stored columns for automatic calculations (salary change percentages and net pay), CHECK constraints enforcing valid enum values and business rules, approval tracking on salary changes and bonuses, soft-delete support for departments, and comprehensive indexing for common query patterns.

Relationships

The schema uses 11 foreign key relationships to enforce referential integrity across all 7 tables. Cascading rules are chosen per-relationship to balance data preservation with cleanup.

Parent	Child	Cardinality	FK column	On delete
departments	employees	1 : N	`department_id`	RESTRICT
employees	departments	1 : 0..1	`manager_id`	SET NULL
employees	salary_history	1 : N	`employee_id`	CASCADE
employees	salary_history	1 : 0..N	`approved_by`	SET NULL
employees	payroll	1 : N	`employee_id`	CASCADE
employees	performance_reviews	1 : N	`employee_id`	CASCADE
employees	performance_reviews	1 : N	`reviewer_id`	SET NULL
employees	bonuses	1 : N	`employee_id`	CASCADE
employees	bonuses	1 : 0..N	`approved_by`	SET NULL
payroll	deductions	1 : N	`payroll_id`	CASCADE
perf_reviews	bonuses	1 : 0..N	`review_id`	SET NULL

Indexes

21 indexes are defined across all tables, optimized for four primary access patterns:

FK lookups — department_id, employee_id, payroll_id, reviewer_id, approved_by. Filtered queries — employment_status, payment_status, deduction_type, bonus_type, review status. Range scans — hire_date, effective_date, review_date, date_awarded, pay periods. Name search — composite index on (last_name, first_name) for directory lookups.

Computed columns

Two columns are defined as GENERATED ALWAYS ... STORED, meaning SQLite calculates and physically stores the values automatically. They cannot be inserted or updated directly.

salary_history.change_percentage — Computes ((new_salary − old_salary) / old_salary) × 100, rounded to 2 decimal places. Returns 0.00 when old_salary is 0 (initial hire).

payroll.net_salary — Computes gross_salary − total_deductions. Stays in sync as deductions are updated.

TABLE 1 OF 7

departments

Core organizational units. Tracks budget, location, and an optional manager reference back to the employees table. Supports soft-delete via the is_active flag to preserve historical data.

7 columns **4** groups

Column	Type	Constraints	Description
IDENTITY			
department_id	INTEGER	PK, AUTO	Unique department identifier
name	TEXT	NOT NULL, UNIQUE	Department name
OWNERSHIP			
manager_id	INTEGER	FK → employees	Department head (nullable)
ATTRIBUTES			
location	TEXT	NOT NULL, DEFAULT 'HQ'	Office location
budget	REAL	NOT NULL, CHECK ≥ 0	Annual budget allocation
is_active	INTEGER	NOT NULL, DEFAULT 1	Soft-delete flag (1 = active)
TIMESTAMPS			
created_at	TEXT	NOT NULL, DEFAULT now	Record creation date

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employees

Central hub of the schema. All downstream tables reference this table via employee_id. Stores personal info, demographics, role, department assignment, pay frequency, and current base salary.

17 columns **5** groups

Column	Type	Constraints	Description
IDENTITY			
employee_id	INTEGER	PK, AUTO	Unique employee identifier
first_name	TEXT	NOT NULL	Legal first name
last_name	TEXT	NOT NULL	Legal last name
email	TEXT	NOT NULL, UNIQUE	Work email address
phone	TEXT	nullable	Contact phone number
DEMOGRAPHICS			
gender	TEXT	CHECK enum	male / female / non_binary / prefer_not_to_say
address	TEXT	nullable	Mailing address and tax jurisdiction
date_of_birth	TEXT	NOT NULL	Format: YYYY-MM-DD
EMPLOYMENT			
hire_date	TEXT	NOT NULL, DEFAULT now	Employment start date
termination_date	TEXT	nullable	Date employment ended (NULL while active)
job_title	TEXT	NOT NULL	Current job title
department_id	INTEGER	NOT NULL, FK → departments	Assigned department
employment_status	TEXT	NOT NULL, CHECK enum	active / on_leave / terminated / probation
COMPENSATION			
pay_frequency	TEXT	NOT NULL, DEFAULT monthly	weekly / biweekly / semi_monthly / monthly
base_salary	REAL	NOT NULL, CHECK > 0	Current annual base salary
TIMESTAMPS			
created_at	TEXT	NOT NULL, DEFAULT now	Record creation timestamp
updated_at	TEXT	NOT NULL, DEFAULT now	Last modification timestamp

TABLE 3 OF 7

salary_history

Immutable audit trail for every compensation change. The change_percentage column is auto-calculated via a generated stored column. Includes approval tracking for compliance.

9 columns **5** groups

Column	Type	Constraints	Description
IDENTITY			
salary_history_id	INTEGER	PK, AUTO	Unique record identifier
employee_id	INTEGER	NOT NULL, FK → employees	Employee reference
SALARY CHANGE			
old_salary	REAL	NOT NULL, CHECK ≥ 0	Previous salary (0 for new hires)
new_salary	REAL	NOT NULL, CHECK > 0	Updated salary amount
effective_date	TEXT	NOT NULL	Date change takes effect
change_reason	TEXT	NOT NULL, CHECK enum	hire / promotion / annual_raise / merit_raise / adjustment / demotion
APPROVAL			
approved_by	INTEGER	FK → employees	Manager who authorized the change
COMPUTED			
change_percentage	REAL	GENERATED (stored)	Auto-calculated: ((new − old) / old) × 100
TIMESTAMPS			
created_at	TEXT	NOT NULL, DEFAULT now	Record creation timestamp

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payroll

One row per employee per pay period. Tracks gross pay, deductions, and the payment lifecycle (pending → processed → paid). Net salary is a generated column computed automatically.

11 columns **5** groups

Column	Type	Constraints	Description
IDENTITY			
payroll_id	INTEGER	PK, AUTO	Unique payroll record
employee_id	INTEGER	NOT NULL, FK → employees	Employee reference
PAY PERIOD			
pay_period_start	TEXT	NOT NULL	Period start date (YYYY-MM-DD)
pay_period_end	TEXT	NOT NULL, CHECK > start	Period end date (YYYY-MM-DD)
AMOUNTS			
gross_salary	REAL	NOT NULL, CHECK > 0	Gross pay for the period
total_deductions	REAL	NOT NULL, DEFAULT 0	Sum of all deductions
net_salary	REAL	GENERATED (stored)	Auto: gross_salary − total_deductions
PAYMENT			
payment_date	TEXT	nullable	Actual payment date (NULL until processed)
payment_method	TEXT	NOT NULL, DEFAULT direct_deposit	direct_deposit / check / wire_transfer
payment_status	TEXT	NOT NULL, CHECK enum	pending / processed / paid / failed / cancelled
TIMESTAMPS			
created_at	TEXT	NOT NULL, DEFAULT now	Record creation timestamp

TABLE 5 OF 7

deductions

Itemized deductions attached to a payroll record. Each row is a single line item. The sum of deductions for a payroll_id should match payroll.total_deductions.

6 columns **3** groups

Column	Type	Constraints	Description
IDENTITY			
deduction_id	INTEGER	PK, AUTO	Unique deduction record
payroll_id	INTEGER	NOT NULL, FK → payroll	Parent payroll record
DEDUCTION DETAIL			
deduction_type	TEXT	NOT NULL, CHECK enum	federal_tax / state_tax / social_security / medicare / health_ins / dental_ins / vision_ins / 401k / hsa / other
amount	REAL	NOT NULL, CHECK > 0	Deduction amount
description	TEXT	nullable	Optional notes
TIMESTAMPS			
created_at	TEXT	NOT NULL, DEFAULT now	Record creation timestamp

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performance_reviews

Periodic evaluations on a 1–5 rating scale. Follows a workflow: draft → submitted → acknowledged. The reviewer is typically the employee's direct manager.

10 columns **4** groups

Column	Type	Constraints	Description
IDENTITY			
review_id	INTEGER	PK, AUTO	Unique review record
employee_id	INTEGER	NOT NULL, FK → employees	Employee being reviewed
reviewer_id	INTEGER	NOT NULL, FK → employees	Manager conducting review
REVIEW DETAILS			
review_date	TEXT	NOT NULL, DEFAULT now	Date of review
review_period	TEXT	NOT NULL, CHECK enum	Q1 / Q2 / Q3 / Q4 / annual / probation
rating	INTEGER	NOT NULL, CHECK 1–5	Performance score
status	TEXT	NOT NULL, DEFAULT draft	draft / submitted / acknowledged
FEEDBACK			
comments	TEXT	nullable	Reviewer's written feedback
goals	TEXT	nullable	Goals for next period
TIMESTAMPS			
created_at	TEXT	NOT NULL, DEFAULT now	Record creation timestamp

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bonuses

One-time or recurring bonus payments. Can optionally link to a performance review via review_id. Includes an approval chain via approved_by for accountability.

9 columns **4** groups

Column	Type	Constraints	Description
IDENTITY			
bonus_id	INTEGER	PK, AUTO	Unique bonus record
employee_id	INTEGER	NOT NULL, FK → employees	Recipient employee
BONUS DETAILS			
amount	REAL	NOT NULL, CHECK > 0	Bonus amount
bonus_type	TEXT	NOT NULL, CHECK enum	performance / signing / referral / holiday / spot / retention / other
date_awarded	TEXT	NOT NULL, DEFAULT now	Date bonus was awarded
reason	TEXT	nullable	Justification for bonus
RELATIONSHIPS			
review_id	INTEGER	FK → performance_reviews	Linked review (optional)
approved_by	INTEGER	FK → employees	Manager who signed off
TIMESTAMPS			
created_at	TEXT	NOT NULL, DEFAULT now	Record creation timestamp